

4 Combinations of random variables	4.1	Distribution of linear combinations of independent Normal random variables.	If $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$ independently, then $aX \pm bY \sim N(a\mu_x \pm b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2)$ No proofs required.
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